

NTSB Bird-Strike Case

NTSB AIR REPORT

REPO ID: CEN25LA070



At night, a Part 135 flight was 5 miles north of its destination at 3,000 ft when it encountered birds. The crew did not see the birds before impact. The bird strike made the captain's primary flight display (PFD) unusable.

extract

EVENT EXTRACTION

ordered evidence units

Nighttime

5 mi · 3,000 ft

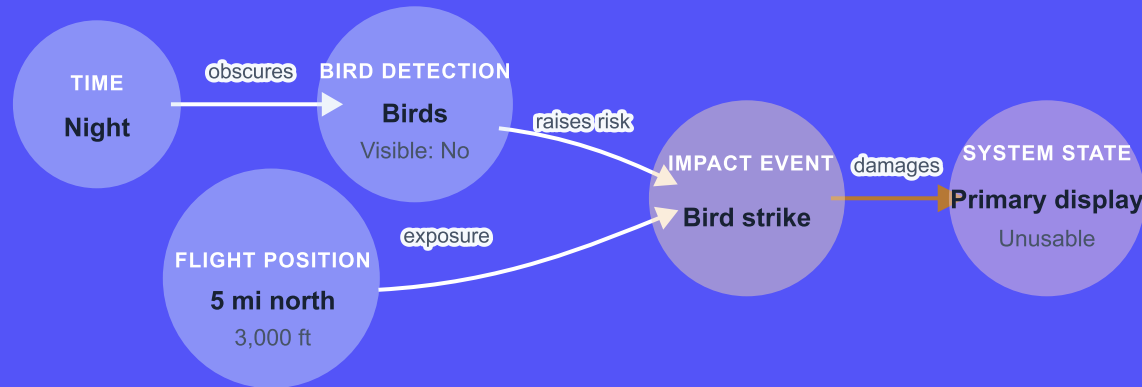
Birds not visible

Bird strike

Primary display failed

map to nodes

Causal Knowledge Graph



DAG definition

$$x_d = [G < t, d_t]$$

Context-derived nodes and edges

DECISION QUESTION

How should the captain respond
after the primary flight display fails?

Task 1 · Possible Actions

Generate options and score each one.



POSSIBLE NEXT ACTIONS

— actual path - - - imagined paths

TASK 1 SCORES

STR SP SC U

A Confirm the remaining
flight instruments.

ACTUAL · O_fact → TASK 2

.69 .35 .50 .39

B Record the strike location
and altitude.

IMAGINED
Spatial-Temporal Reconstruction
STR

.85 .47 .36 .49

C Shift primary monitoring
to the first officer.

IMAGINED
Self Projection(SP)

.40 .70 .45 .31

D Notify ATC of the
display failure.

IMAGINED
Self Consistency(SC)

.79 .35 .42 .24

DECISION POINT

 d_t

WHAT NEXT?

choose an action



TASK 1 EVALUATION TARGET

Option × dimension score matrix

$$S_{Task1} \in [0, 1]^{(4 \times 4)}$$

Shown values are benchmark targets for Task 1

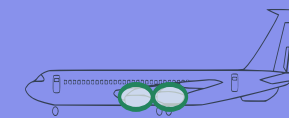
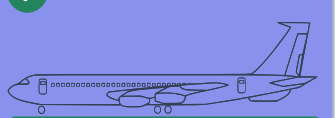
Task 2 · Actual Outcome

FACTUAL ACTION FROM OPTION A

A Confirm the remaining
flight instruments.

O_fact

WHAT WILL HAPPENED NEXT ?

first-officer display
+ standby availableengines
unaffectedlanded without
further incident

no injuries

damage
forward-fuselage

OUTCOME LABELS

GA · RE · CV · SEC

outcome tags: safe landing · damage · resource change